

RepoScope

See your repo. Understand it. Track it.

Built with IBM Bob + Z Code · IBM Bob 2.0 Hackathon · lablab.ai · 2026

Understanding a repo is broken.



GitDiagram — renders the graph, doesn't answer questions.



GitReverse — generates a prompt, doesn't show structure.



Gitingest — dumps text, no visualization.



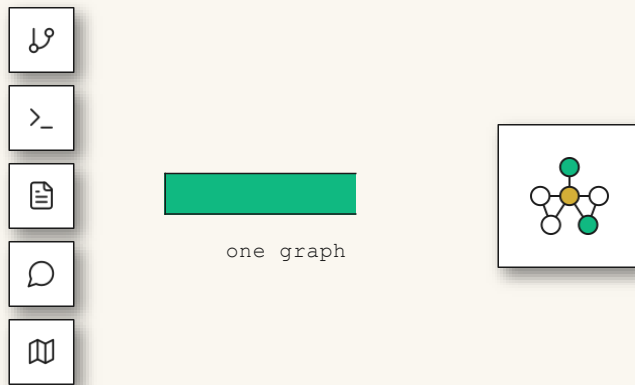
Explain GitHub — Q&A, no tour, no tracking.



Tour tools — guided walkthroughs, disconnected from everything.

Five tabs. Five mental models. Zero cohesion.

Five tools become one.



Every one of those tools needs the same thing: a knowledge graph of the repo — files, functions, imports, commits, PRs, issues, owners, incidents.

Build that graph once. Then every feature is a view on top.

The graph is the product. Everything else is a lens.

RepoScope — the workspace for any GitHub repo.



Ask.

Natural-language Q&A with file, function, and commit citations.

See.

Interactive architecture graph that groups nodes by layer.

Track.

Commits, PRs, issues, and releases — checked on demand.

What you get on day one.



Architecture Graph

Interactive layered diagram — client, backend, storage, external, compute.



Conversational Q&A

Ask anything. Answers cite files, functions, and commit SHAs.



Code Tours

Guided walkthroughs of any module, step by step.



Impact Analysis

"What breaks if I change this?" — with blast radius on the graph.



Reverse Build Prompt

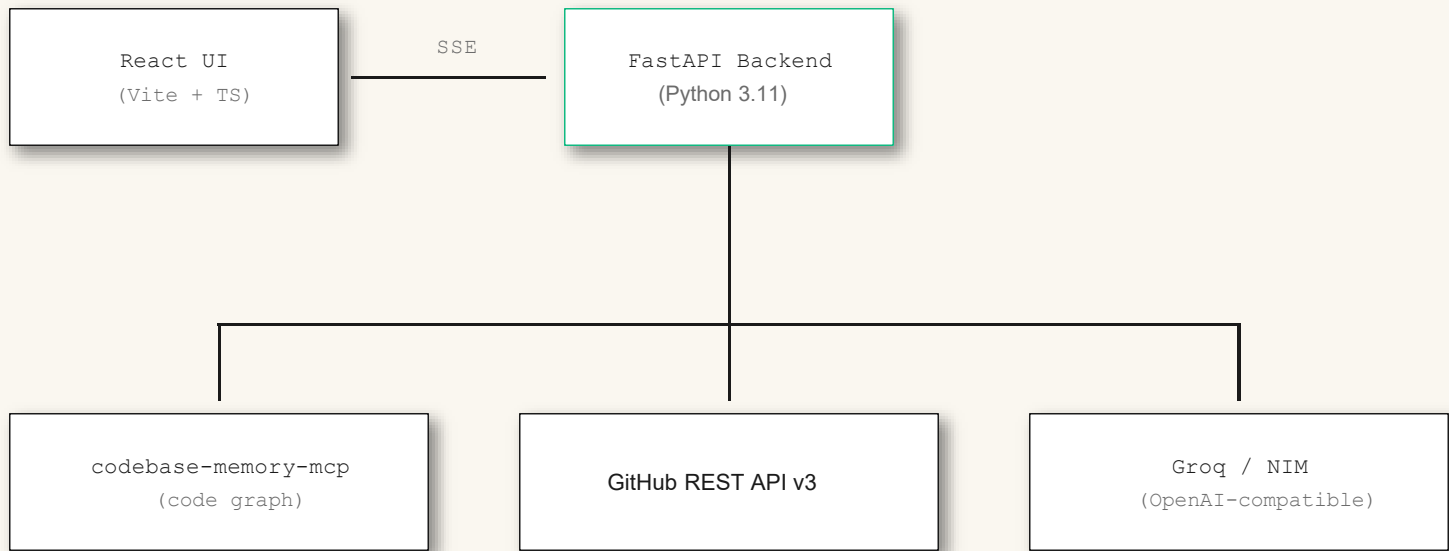
Turn any repo into an agent-ready prompt that rebuilds it.



BYOK + Tracking

Paste your GitHub PAT to access private repos. Track commits, PRs, and releases on demand.

Built on a single knowledge graph.



■ Graph built once at repo load.

■ Every feature is a query on top.

■ Groq primary, NVIDIA NIM fallback.

Five tools vs. one workspace.

Capability	The 5 tools	RepoScope
Architecture graph	GitDiagram only	✓ Built in
Reverse prompt	GitReverse only	✓ Built in
Repo summary	Gitingest only	✓ Built in
Chat Q&A	Explain GitHub	✓ Built in
Guided tours	Separate tool	✓ Built in
Track changes	None	✓ Built in
Impact analysis	None	✓ Built in
Private repos	Paywalls	✓ BYOK

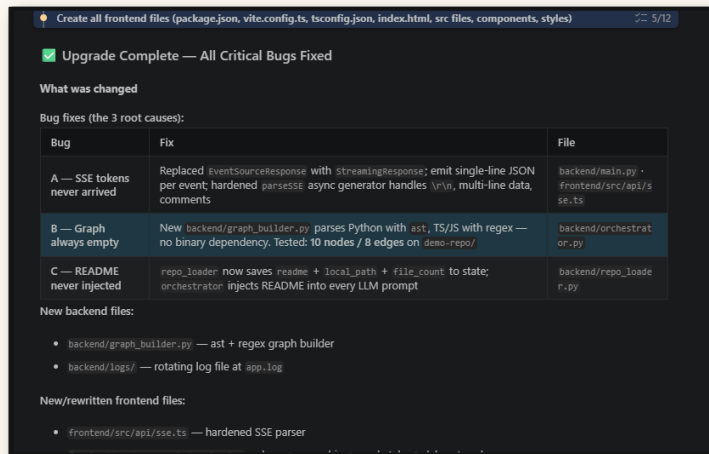
Every other tool does one thing. RepoScope does all of them, on one graph.
graph.

Built with IBM Bob + Z Code

HOW WE USED BOB

- **Architecture design** — Bob drafted the module layout.
- **Backend** — Bob wrote FastAPI endpoints, MCP client, and the graph builder.
- **Frontend** — Bob wrote the React components, the graph renderer, and the SSE client.
- **Hardening** — Bob added error boundaries, race guards, and validation.
- **Documentation** — Bob wrote every markdown file in the repo.

PROOF



Most tasks ran through IBM Bob; Z Code (GLM) paired on the complex ones — graph layout, hardening, and this deck.

What it looks like in action.

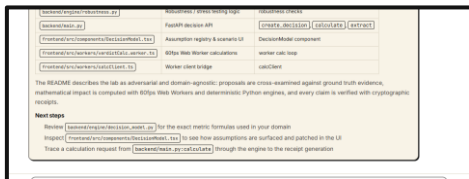
01



Paste a URL. The graph renders in seconds.



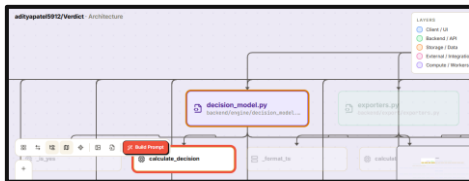
02



Ask 'How does auth work?' — get cited answers.



03



Change a function. See the blast radius.



Try RepoScope

See your repo. Understand it. Track it.



GitHub

github.com/adityapatel15912/RepoScope



Live demo

repo-scope-nu.vercel.app



Video

youtu.be/ZJ7PMCZRXY